Payments Optimization: Where the Margin Leaks and Which Fixes Pay Back

An end-to-end analytics project on a fictional EU payments platform: raw payment data modelled in BigQuery and dbt, tested at every layer, and turned into a ranked list of fixes valued in euros per year. Every leak is measured by shop and month, and every bank payout is checked against the payments behind it. Python simulated data of 12 months.

Month, Year of Paymen..
- ✔ January 2025
- ✔ February 2025
- ✔ March 2025
- ✔ April 2025
- ✔ May 2025
- ✔ June 2025
- ✔ July 2025
- ✔ August 2025
- ✔ September 2025
- ✔ October 2025
- ✔ November 2025
- ✔ December 2025

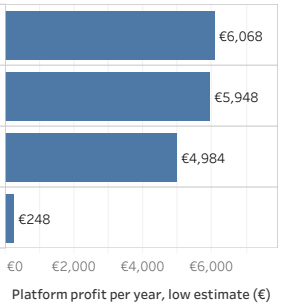
Business Type
- ✔ Digital subscriptions
- ✔ Food delivery
- ✔ Online retail
- ✔ Travel

Totals, 2025

Sales	€11,521,156
Sales Lost (Declines + Give-Ups)	€2,340,290
Platform Revenue	€230,790
Bank Fees	€176,496
Fraud Cost (Paid by platform)	€11,700
Platform Margin	€42,594

Optimizations ranked by the low estimate of platform profit, yearly

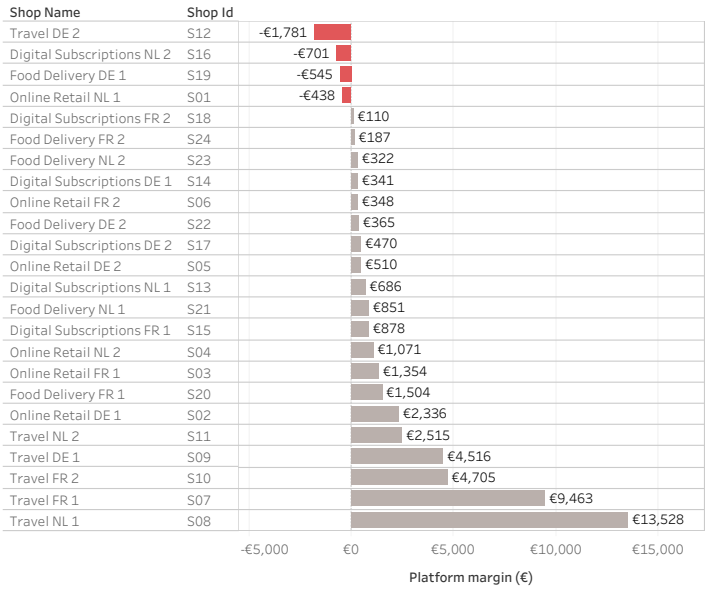
Rank	Optimization Action	Change in Shop Sales	Platform Profit Estimate
1	Raise shop fees to a safe level for shops on which the platform loses money	€0	€6,068
2	Send payments to a bank in the card's own country when available.	€144,378	€5,948
3	Enforce 3D Secure on every payment to remove the platform's fraud costs	-€214,555	€4,984
4	Turn on automatic retries for declined payments at shops that have never used them	€44,639	€248



Flagged shops: current fee, safe fee and 2025 platform margin

Flag	Shop Name	Shop Id	Platform Margin	Current Fee	Safe Fee
Loses money: raise fee to the safe level	Travel DE 2	S12	-€1,781	1.50%	2.28%
	Digital Subscriptions NL 2	S16	-€701	1.50%	4.53%
	Food Delivery DE 1	S19	-€545	0.90%	1.15%
	Online Retail NL 1	S01	-€438	1.10%	1.20%
More fraud than its business type	Digital Subscriptions DE 1	S14	€341	2.40%	
Loses money only if fraud runs high	Digital Subscriptions NL 1	S13	€686	2.40%	

Platform margin per shop, 2025 (red = losing money)



Bank payouts that don't match the payments, 2025

Shop Name	Shop Id	Bank Id	Missing payouts	Not paid out	Wrong-fee payouts	Over Charged	To recover
Travel FR 1	S07	BNK_DE	1	€14,704	0	€0.00	€14,704
Online Retail NL 1	S01	BNK_NL	1	€2,265	11	€37.60	€2,302
Online Retail FR 2	S06	BNK_FR	1	€234	0	€0.00	€234
Digital Subscriptions DE 1	S14	BNK_FR	1	€64	0	€0.00	€64
Travel NL 1	S08	BNK_NL	0	€0	15	€18.89	€19
Food Delivery DE 1	S19	BNK_NL	0	€0	13	€11.07	€11
Food Delivery NL 1	S21	BNK_NL	0	€0	21	€9.26	€9
Online Retail NL 2	S04	BNK_NL	0	€0	15	€6.66	€7
Digital Subscriptions NL 1	S13	BNK_NL	0	€0	7	€3.24	€3
Food Delivery NL 2	S23	BNK_NL	0	€0	14	€3.07	€3

Code, models, tests and full method:

https://github.com/a-mc-devops/payments_optimization

Data: simulated data for a fictional EU payments platform: 24 shops, 3 banks, 6 card countries and about 100,000 payments from January to December 2025. A Python generator built the data from a written rulebook, with planted problems such as underpriced shops, wrong bank fees and missing payouts. The answers were locked away before the analysis began: only a fingerprint of the answer file was saved, so the results could not be tuned to them.

Pipeline: six raw tables loaded into BigQuery and modelled with dbt in three layers (staging, intermediate, marts): 18 models checked by 135 tests, from unique keys to totals that must match their inputs to the cent. 317 duplicate payment rows were removed in staging. The dashboard reads only the final tables.

Validation: after the analysis, the results were compared with the locked answers. Every planted problem was found, and the fraud cost and the platform margin came within 3% and 1% of their true values.

Platform margin = platform revenue (the price shops pay) − bank fees − fraud cost. **Fraud cost** is what the platform pays when fraud happens on a payment without 3D Secure: the amount back to the card owner plus a €20 dispute fee, both deducted from the platform margin.

Optimizations are valued one at a time over 2025 and cannot be added together. Each one is valued twice, in platform profit and in shop sales, because some fixes trade one for the other. The estimate ranges are built using a $\pm 2\sqrt{k}$ deviation, where k is the number of fraud cases. Optimizations are ranked by the low estimate, the cautious value. Options that lose money even at the high estimate are left out. The ranking covers all shops and does not follow the filters.

Change in shop sales: a statistical estimate, based on the available data, of how much more or less the shops would sell in a year with the fix in place.

Safe fee: the threshold fee at which the platform still breaks even on the shop when fraud runs at the high estimate.

Payout checks: each bank payout is matched and compared with the approved payments it should contain, per shop, bank and day. Mismatches and missing payouts are listed in the table for further inspection.